

# Aurick Anwar

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## EDUCATION

### McMaster University

Bachelor of Engineering: Engineering Physics

Hamilton, Ontario

*Expected 2029*

## WORK EXPERIENCE

### Software Engineer

McMaster Drone Team

Hamilton, Ontario

*June 2026 – Present*

- Developed an autonomous drone pipeline using **YOLOv11**, **OpenCV**, and **PyTorch**, achieving **30+ FPS** real-time object detection with **90%+ mAP** in **ArduPilot** simulated environments.
- Simulated and validated **100+** autonomous flight missions in **NVIDIA Isaac Sim**, testing navigation and obstacle avoidance in different environments before deploying.
- Built **10+ ROS2** nodes, enabling a **<100 ms** communication between perception, planning, and control systems.

### Software Engineering Intern

HermesAI

Toronto, Ontario

*May 2026 – August 2026*

- Built **React** interfaces for document upload and AI-powered case extraction, supporting legal files exceeding **1,000 pages**.
- Built an **LLM** chatbot feature with **LangChain** that parses, chunks, and extracts case law from legal documents, cutting manual processing time by **50%** for Canadian Law Firms.
- Reduced document processing latency by **30%** by offloading file processing to asynchronous **Node.js**.

## PERSONAL PROJECTS

### 🔗 TeleARM

[ROS, SLAM, Gazebo, LiDAR, RGB Camera, FoxGlove]

- Built an autonomous **ROS2** pipeline that uses an **RGB Camera** to detect colored objects from the ground and triggers a forklift-style pickup, reducing manual clean-up by **25%**.
- Used **SLAM + Nav2** stack in **Gazebo**, letting the robot re-route with a **LiDAR** around walls and land at the correct route markers, improving navigation reliability by **50%**.
- Designed a modular system separating perception, navigation, and manipulation into independent **ROS2** nodes, visualized in **Foxglove**, cutting debugging time by **40%**.

### 🔗 Autonomous Self-Driving Vehicle with CARLA

[CARLA, YOLOv11, PyTorch, OpenCV]

- Developed an end-to-end autonomous driving pipeline in the **CARLA** simulator integrating **YOLO-based object detection** and vehicle control, achieving a real-time simulation at **~20-30 FPS**.
- Wrote to a dataset with **5+ features** (steer, brake, speed, acceleration, throttle), generating **25k+** sample datasets to train a **PyTorch** model.
- Used **CrossEntropyLoss**, achieving **~88.5%** prediction directional accuracy for precise automation navigation.

### 🔗 Breast Cancer Cell Detection

[PyTorch, Roboflow, YOLOv11, Grad-Cam, CNNs]

- Fine-tuned **Roboflow YOLOv11** tech stack on **5,400+** images to localize breast cancer regions with bounding box detection across **benign** and **malignant** classes.
- Built a custom **3-layer PyTorch CNN** classifier achieving **95.31%** confidence on malignant tumor classifications using **CrossEntropyLoss** and **Adam optimization**.
- Implemented **Grad-CAM**, generating **spatial heatmaps** that visually highlight the tumor regions driving each prediction.

## Technical Skills

**Programming Languages:** Python, C++, Javascript, CSS, HTML, MATLAB,

**ML and Computer Vision:** PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLOv11, LangChain, Mediapipe

**Robotics and Simulation:** ROS, RVIZ, Gazebo, SLAM, Nvidia Isaac Sim, MuJoCo, ArduPilot, FoxGlove,

**Electronics and Microcontrollers:** Arduino, Raspberry Pi, ESP32

**3D Design and CAD:** Solidworks, Inventor, Fusion360, AutoCAD

**Web Development:** Node.js, React

**Tools and OS:** Linux, Git, Docker